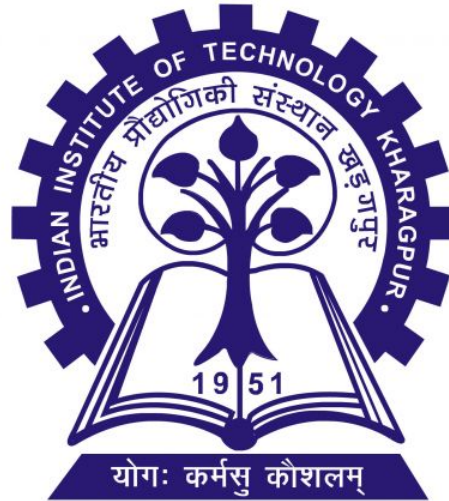


VLSI DESIGN LAB [EC69216]



EXPERIMENT - 5

Design of 4-bit Adder Pipeline Unit

Group No. 4

- **Member 1:** Arup Basak 22E37021
- **Member 2:** Chiradip Biswas 22EC37004
- **Group no.:** 4
- **Date of Submission:** 16-03-2026

Aim

- Design and simulate 1-bit full-adder circuit. Verify its functionality and observe its critical path delays (both, worst case and best case) with load capacitance equal to input capacitance of four unit size inverters together.
- Make a DRC and LVS clean Physical design of the full-adder circuit. Verify and compare its pre- layout and post-layout performance.
- Design and simulate a 4-bit adder using the designed 1-bit full-adder in part (a) of this experiment.
- Construct parallel-in-parallel-out 4-bit Register using the D-FF (the D-FF obtained in experiment 3 or, some other alternative circuit may be used).
- In simulation, demonstrate maximum speed of operation of a pipeline unit containing the 4-bit adder as combinational circuit in-between two register layers (one layer is for feeding the inputs and the other one is for storing the output)

1 Circuits and Layouts:

1.1 NAND gates

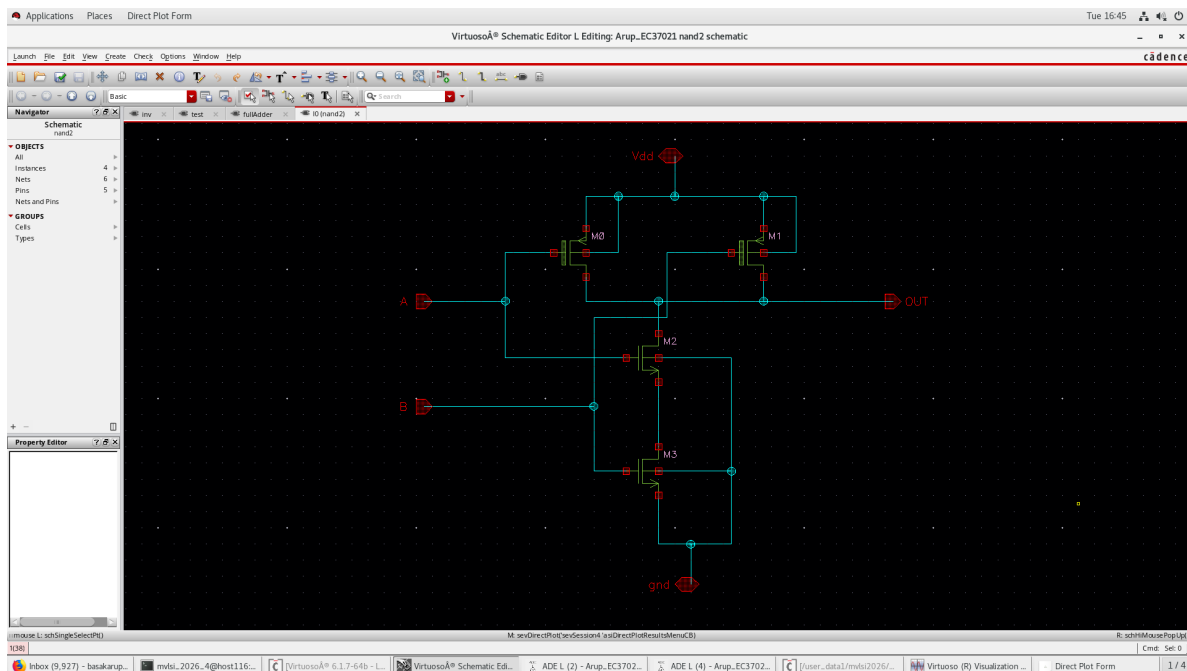


Figure 1: nand gate circuit [Wp=6u Wn=4u]

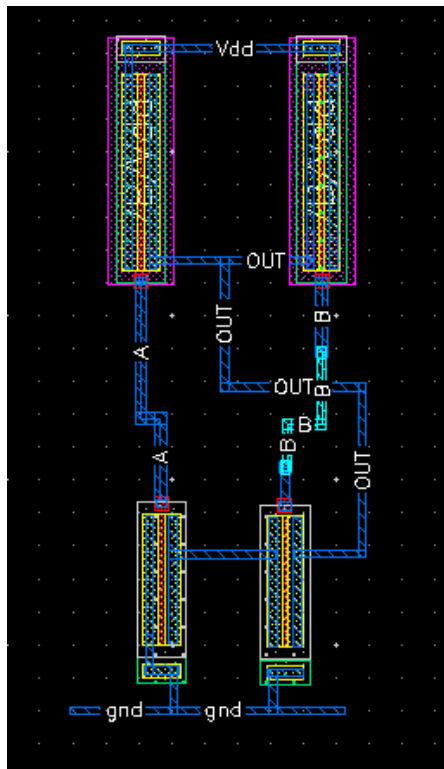


Figure 2: nand layout

1.2 Full Adder

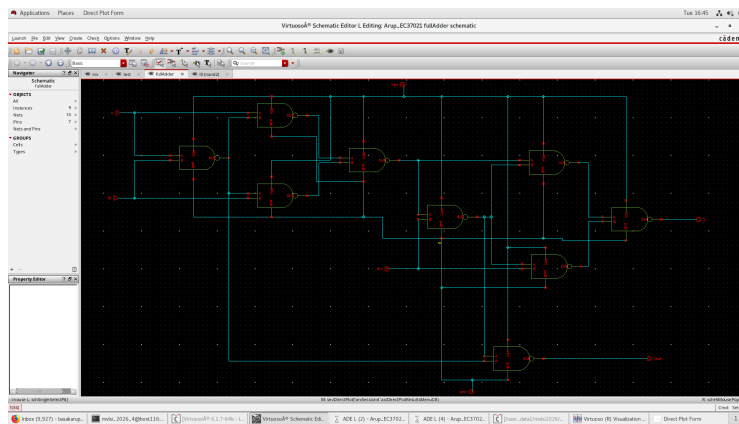


Figure 3: full adder circuit

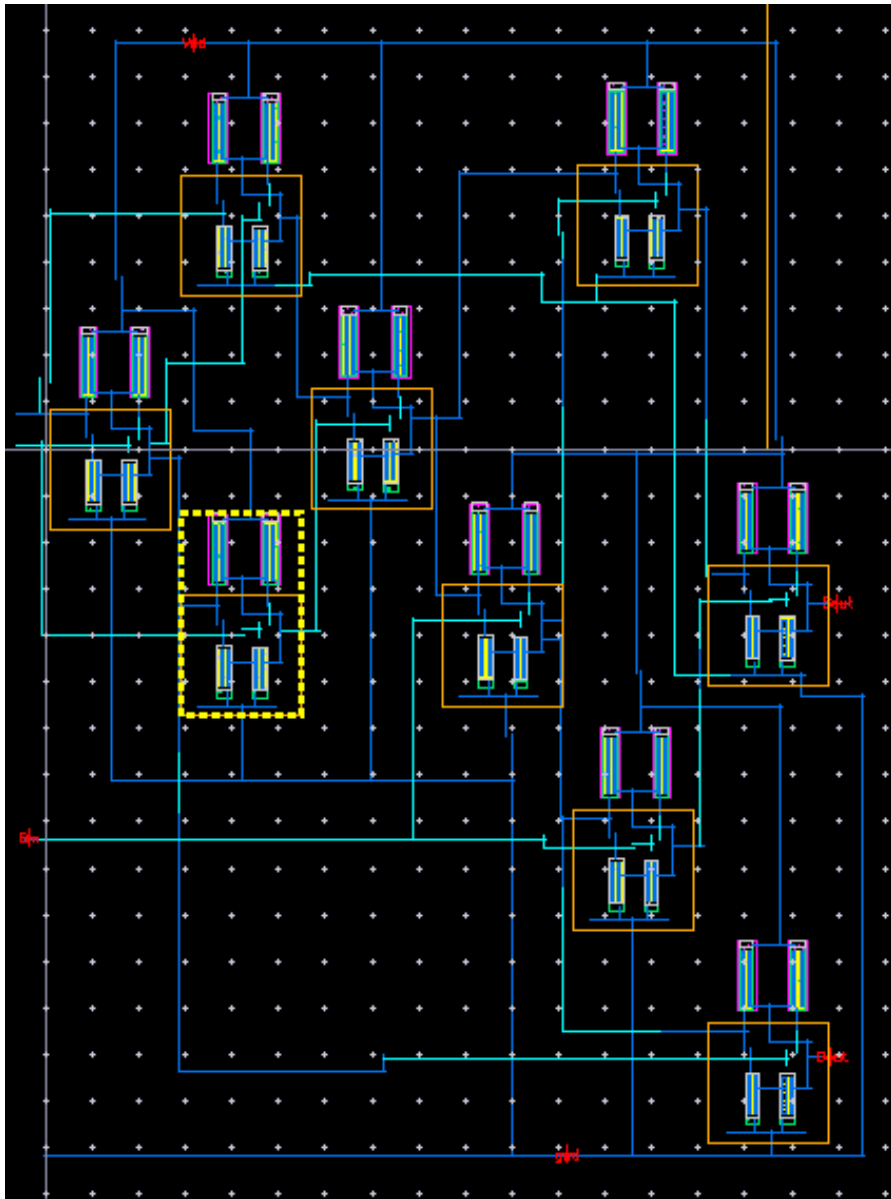


Figure 4: full adder layout layout

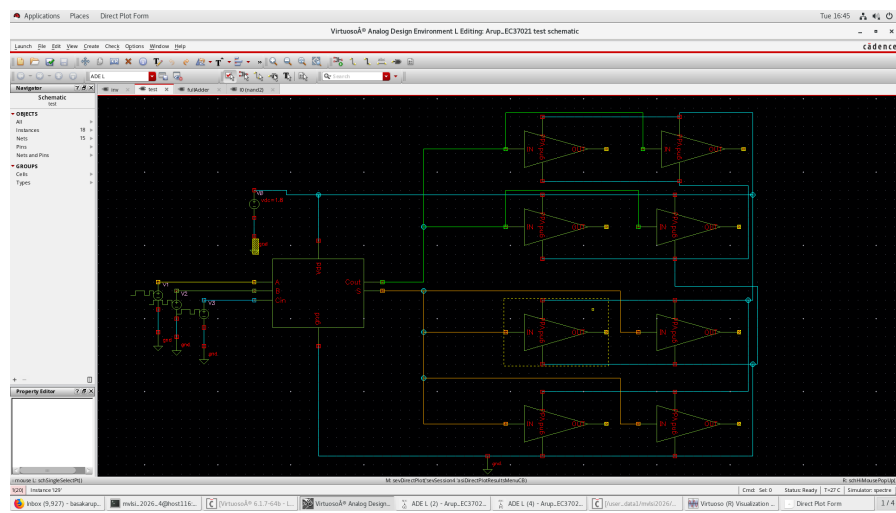


Figure 5: full adder driving a fanout of 4

1.3 4 bit Full Adder

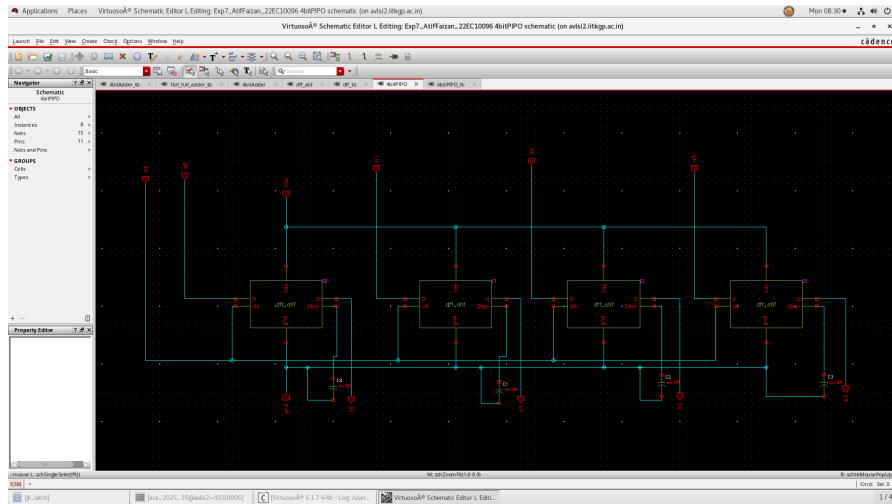


Figure 6: 4 bit full adder

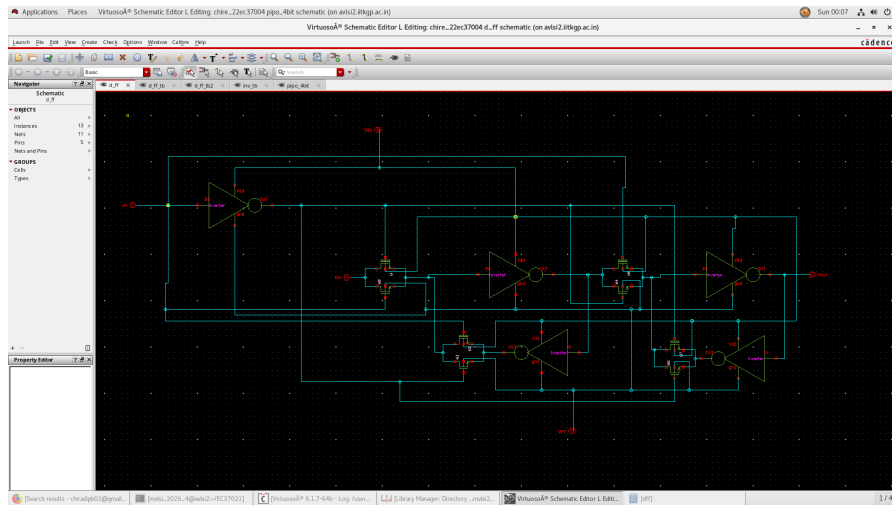


Figure 7: d flip flop using transmission gate

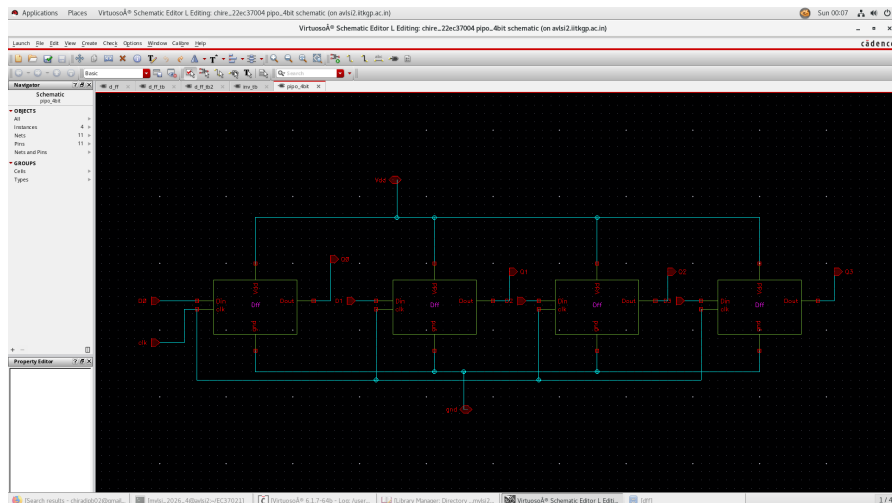


Figure 8: 4 bit Pipo schematic

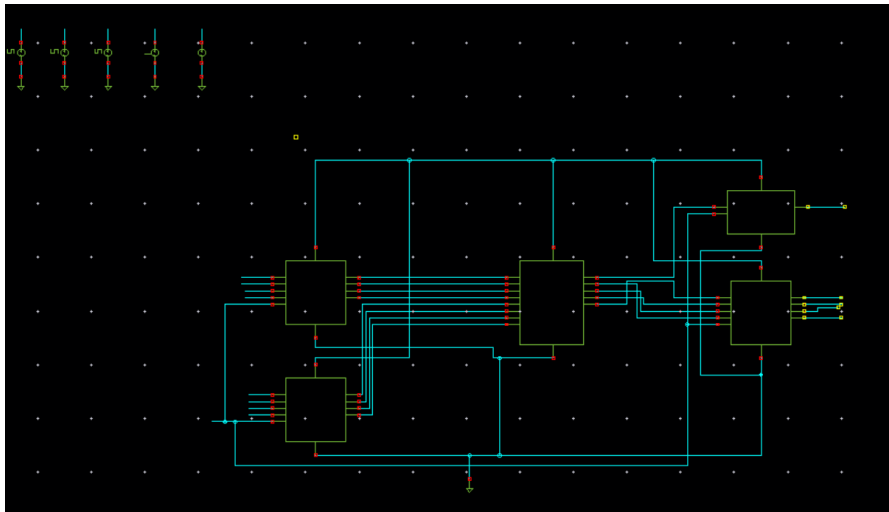


Figure 9: pipeline unit 4 bit adder

2 Observations:

2.1 Full adder

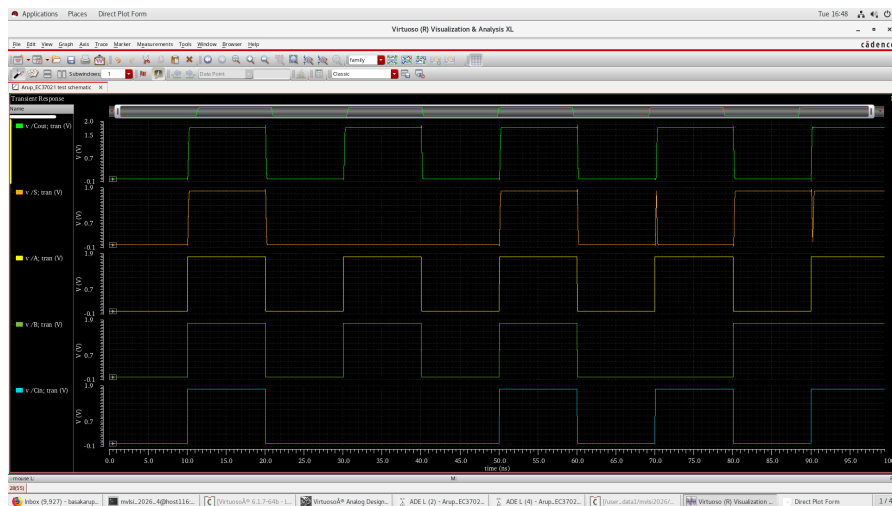


Figure 10: The schematic simulation

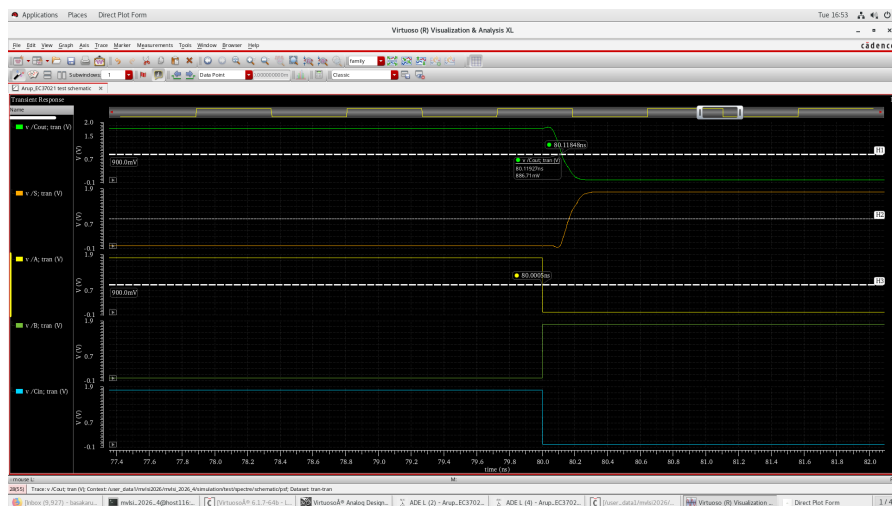


Figure 11: Worst case fall time Cout: 0.127ns

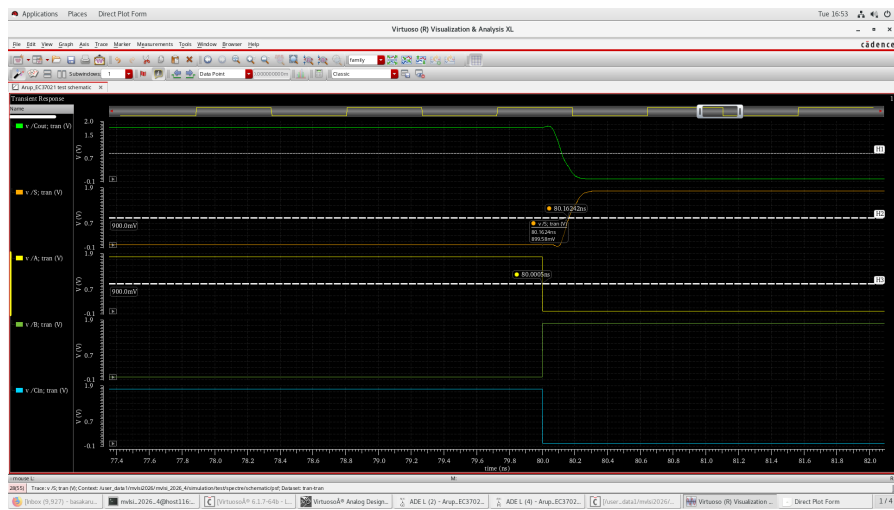


Figure 12: Worst case rise time Sout: 0.162ns

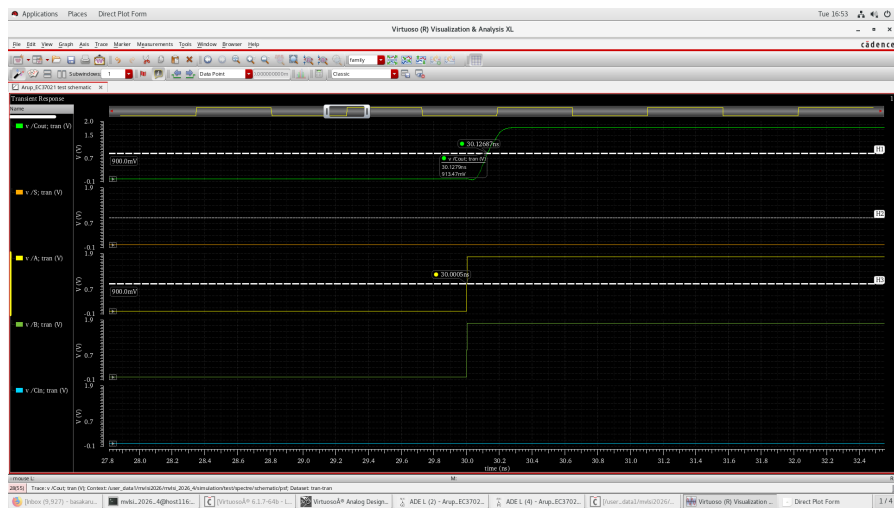


Figure 13: Worst case rise time Cout: 0.206ns

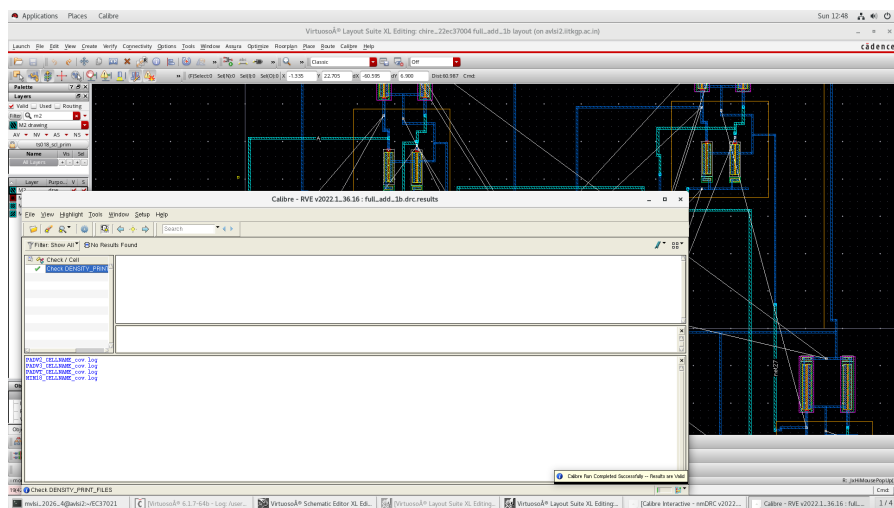


Figure 14: DRC passing of Full adder Layout

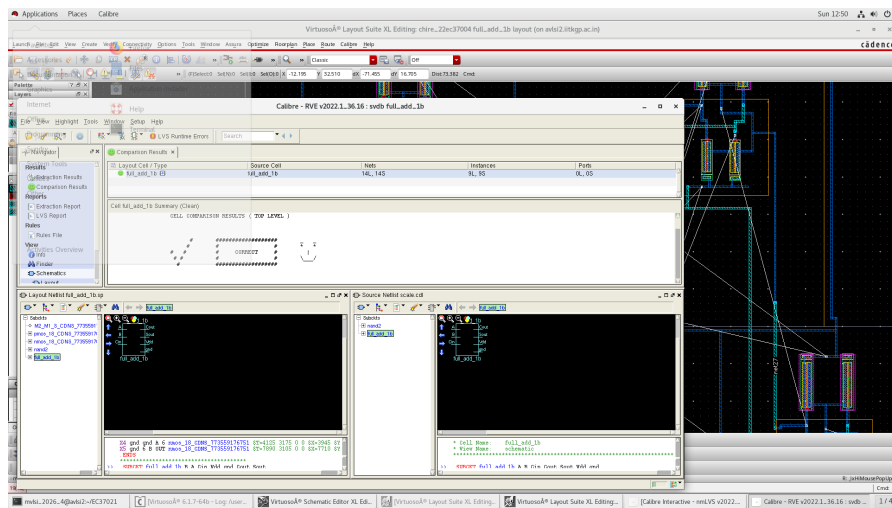


Figure 15: LVS passing of the full adder layout

```

Open [icon] full_add_1b.pex.netlist
~EC37021/chire_22ec37004/full_add_1b/PEX

// File: full_add_1b.pex.netlist
// Created: Sun Mar 15 12:53:48 2026
// Program "Calibre xRC"
// Version "v2022.1.36.16"
//
simulator lang=spectre
include "full_add_1b.pex.netlist.pex"
subckt full_add_1b ( B A Cin Vdd gnd Cout Sout )
//
// SOUT SOUT
// COUT COUT
// GND GND
// VDD VDD
// CIN CIN
// A A
// B B
mXI9/MM0 ( N_net8_XI9/MM0 d N A XI9/MM0 g N Vdd XI9/MM0 s N Vdd XI9/MM0 b \
) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 pd=1.296e-05 ps=1.296e-05
mXI9/MM1 ( N_net8_XI9/MM1 d N B XI9/MM1 g N Vdd XI9/MM1 s N Vdd XI9/MM1 b \
) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 pd=1.296e-05 ps=1.296e-05
mXI9/MM3 ( N_XI9/net50_XI9/MM3 d N A XI9/MM3 g N gnd XI9/MM3 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI9/MM2 ( N_net8_XI9/MM2 d N B XI9/MM2 g N XI9/net50_XI9/MM2 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI10/MM1 ( N_net23_XI10/MM1 d N_net8_XI10/MM1 g N Vdd XI10/MM1 s \
N Vdd XI10/MM1 b ) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 \
pd=1.296e-05 ps=1.296e-05
mXI10/MM0 ( N_net23_XI10/MM0 d N B XI10/MM0 g N Vdd XI10/MM0 s \
N Vdd XI10/MM0 b ) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 \
pd=1.296e-05 ps=1.296e-05
mXI10/MM3 ( N_XI10/net50_XI10/MM3 d N_net8_XI10/MM3 g N gnd XI10/MM3 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI10/MM2 ( N_net23_XI10/MM2 d N B XI10/MM2 g N XI10/net50_XI10/MM2 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI12/MM1 ( N_net22_XI12/MM1 d N A XI12/MM1 g N Vdd XI12/MM1 s \
N Vdd XI12/MM1 b ) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 \
pd=1.296e-05 ps=1.296e-05
mXI12/MM0 ( N_net22_XI12/MM0 d N_net8_XI12/MM0 g N Vdd XI12/MM0 s \
N Vdd XI12/MM0 b ) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 \
pd=1.296e-05 ps=1.296e-05
mXI12/MM3 ( N_XI12/net50_XI12/MM3 d N A XI12/MM3 g N gnd XI12/MM3 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI12/MM2 ( N_net22_XI12/MM2 d N_net8_XI12/MM2 g N XI12/net50_XI12/MM2 s \
N_gnd XI9/MM3 b ) N18 l=1.8e-07 w=4e-06 ad=1.92e-12 as=1.92e-12 pd=8.96e-06 \
ps=8.96e-06
mXI11/MM1 ( N_net13_XI11/MM1 d N_net22_XI11/MM1 g N Vdd XI11/MM1 s \
N Vdd XI11/MM1 b ) P18 l=1.8e-07 w=6e-06 ad=2.88e-12 as=2.88e-12 \

```

Figure 16: After performing PEX the netlist of full adder

2.2 D flip flop

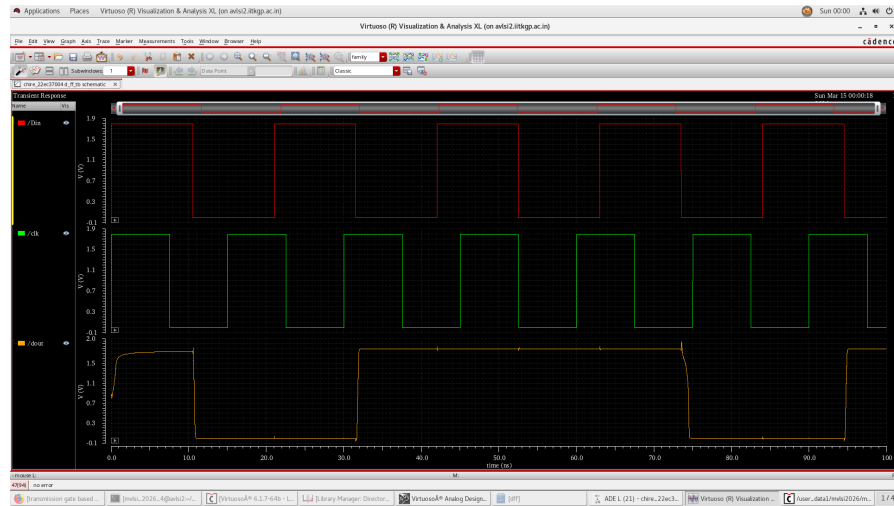


Figure 17: Dff waveform

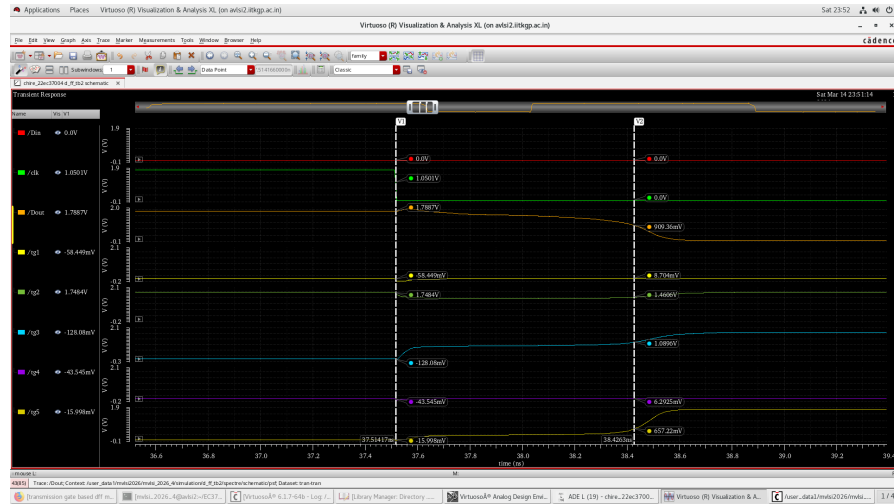


Figure 18: DFF fall time:0.9ns



Figure 19: dff rise time delay:0.5ns

2.3 4 bit adder Pipeline

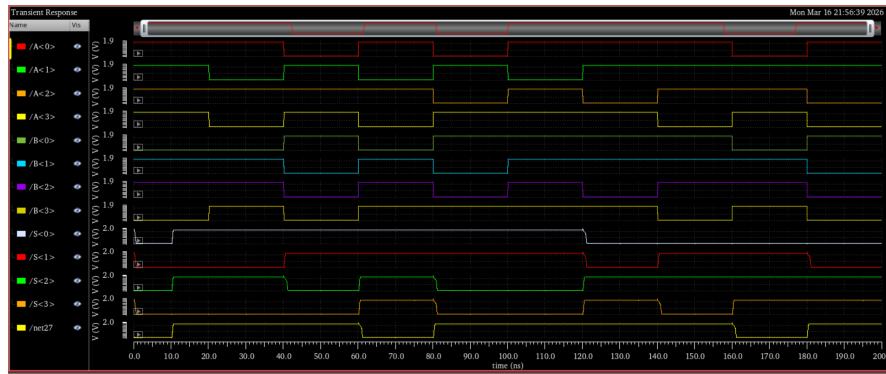


Figure 20: Normal working with 50Mhz clk

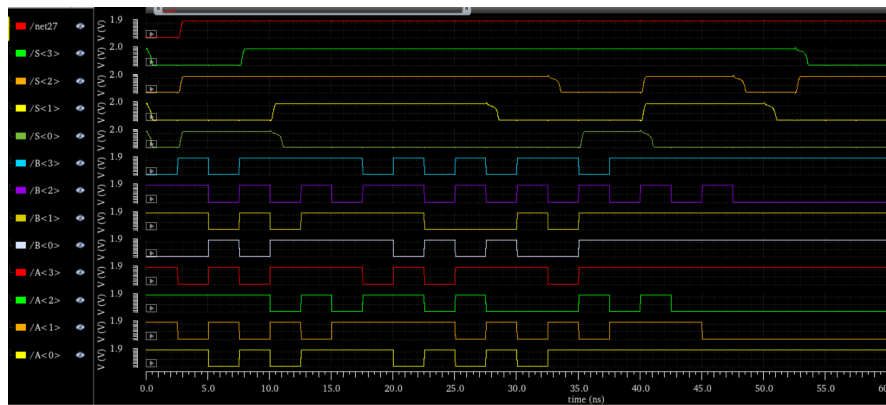


Figure 21: Limiting clk frequency at 500Mhz clk

Discussion Chiradip Biswas 22EC37004

- In this experiment we designed 1 bit Adder (using nand gates), 4 bit Adder, 4 bit PIPO Register and a Pipeline Unit for Addition.
- We used only NAND gates for designing the 1 bit adder and hence we required 9 NAND Gates for the design. The complete design logic for the implementation of 1 bit adder using NAND gates is given in the Logic and Design section. We utilized various laws (like Demorgan's Laws) for simplification of boolean expression for obtaining the simplified version of implementation of adder using NAND gates.
- If we have designed a CMOS based implementation we would have required lesser number of gates.
- For obtaining worst case and best case delay we tried for every possible combinations of A,B and Cin. We obtained worst case delay to be 206ps, from the rising time of Cout. We tested our adder against a load capacitance of 4 times the capacitance of the unit size inverters that we made.
- We made the PIPO using D flip flops. The D flip flops are made using transmission gate and Inverter topology.
- The d flip flops have a rising and falling time delay of 0.9ns, 0.5ns respectively.
- For layout, we used layoutXL and calibre in Cadence. For a successfull layout first it has to pass all the DRC (design rule check), checking all the MOSIS rules of metal layers separation, widths. Then it is to be checked whether the layout gives same circuit as the schematic, for proper operation.
- After the DRC and LVS check passes, PEX (parasitic cap extraction) is to be done, to observe magnitude of all the parasitic capacitances formed (metal-oxide-metal).

- In the layout we generate all components and pins from the schematic. First made the nand gate's layout, using guard ring around nmos and pmos to prevent cmos latchup. Wherever 2 metal wires were crossing, 2 different metal lines are used, with m2-m1 via.
- There are 7 metal layers in total with increasing thickness as increasing in metal number. It is a good practice not to use many metal layers and via as it increases parasitic resistance and capacitance along with size.

Discussion Arup Basak 22EC37021

- The primary objective of this laboratory exercise was to implement and analyze a 1-bit adder, a 4-bit adder, a 4-bit PIPO (Parallel-In Parallel-Out) register, and a pipeline unit dedicated to addition.
- The 4-bit PIPO register was constructed using D flip-flops, which we modeled using a transmission gate and inverter topology.
- For the fundamental 1-bit adder block, the design was restricted exclusively to universal NAND gates. By applying boolean simplification techniques such as De Morgan's laws (detailed in the Logic and Design section), we successfully implemented the adder using exactly 9 NAND gates. It is worth noting that a standard CMOS-based logic design would have yielded a more hardware-efficient circuit with fewer overall gates.
- Timing analysis was conducted by testing all possible input transitions across A, B, and Cin to capture the absolute performance limits. Driving a capacitive load equivalent to four of our unit-sized inverters, the circuit demonstrated a best-case delay of 186.4 ps and a critical worst-case delay of 191.72 ps.
- The physical layout phase was executed using Cadence LayoutXL and Calibre. The components and pins were instantiated directly from our schematic. A critical part of the physical design involved placing guard rings around the PMOS and NMOS devices in the NAND gates to prevent CMOS latch-up.
- To ensure manufacturability and functional correctness, the design had to clear Design Rule Checks (DRC)—verifying MOSIS constraints like metal widths and spacing—followed by Layout Versus Schematic (LVS) to guarantee the layout accurately matched the intended circuit diagram.
- After achieving a clean DRC and LVS, we performed Parasitic Extraction (PEX). This step is crucial for estimating the physical metal-oxide-metal parasitic capacitances introduced by the layout geometry.
- During routing, overlapping signal paths were resolved by utilizing different metal layers connected via contacts (e.g., M1-M2 vias). Although the process supports up to 7 distinct metal layers of increasing thickness, we intentionally minimized the use of higher-level metals and excessive vias to keep the footprint compact and reduce parasitic resistance and capacitance.